$H \rightarrow WW$ + charm

Analysis status

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2022postEE · 26.7 fb^{-1} · expected UL 1034

Sections

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Dedicated decks: `ctag-sf-deck.pdf` (34 slides) · `negrw-deck.pdf` (57 slides)

1 · Analysis and selection

Signal and final state

$H \to WW \to 2\ell 2\nu$, produced with a charm quark.

Final state: **opposite-sign eμ** + MET + **≥1 c-tagged jet**

object	selection
muons	tight ID + iso, $p_T > 10$, $\lvert\eta\rvert < 2.4$
electrons	wp80iso, $p_T > 10$, $\lvert\eta\rvert < 2.5$
ll pair	OS, $p_T > 20 / 10$, $m_{ll} > 12$
c-jets	$p_T > 20$, PNet medium CvL/CvB

> eμ removes the Z peak by construction — the dominant background is **real $t\bar{t}$**.

Data: ReReco `22Sep2023`, all eras.

2 · MVA-defined regions

Six classes, one network

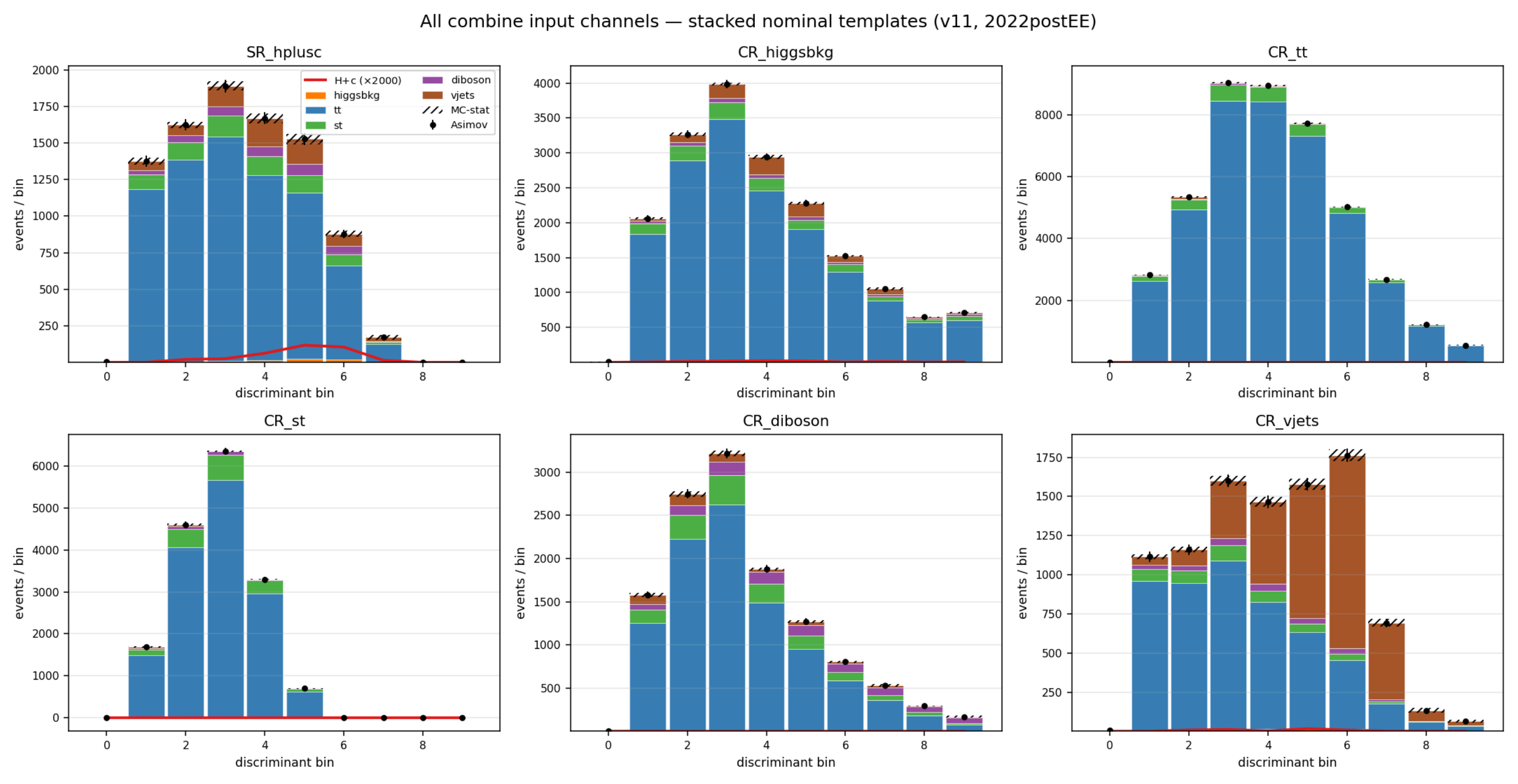
`[hplusc, higgsbkg, tt, st, diboson, vjets]`

argmax defines the region · the winning score is the discriminant

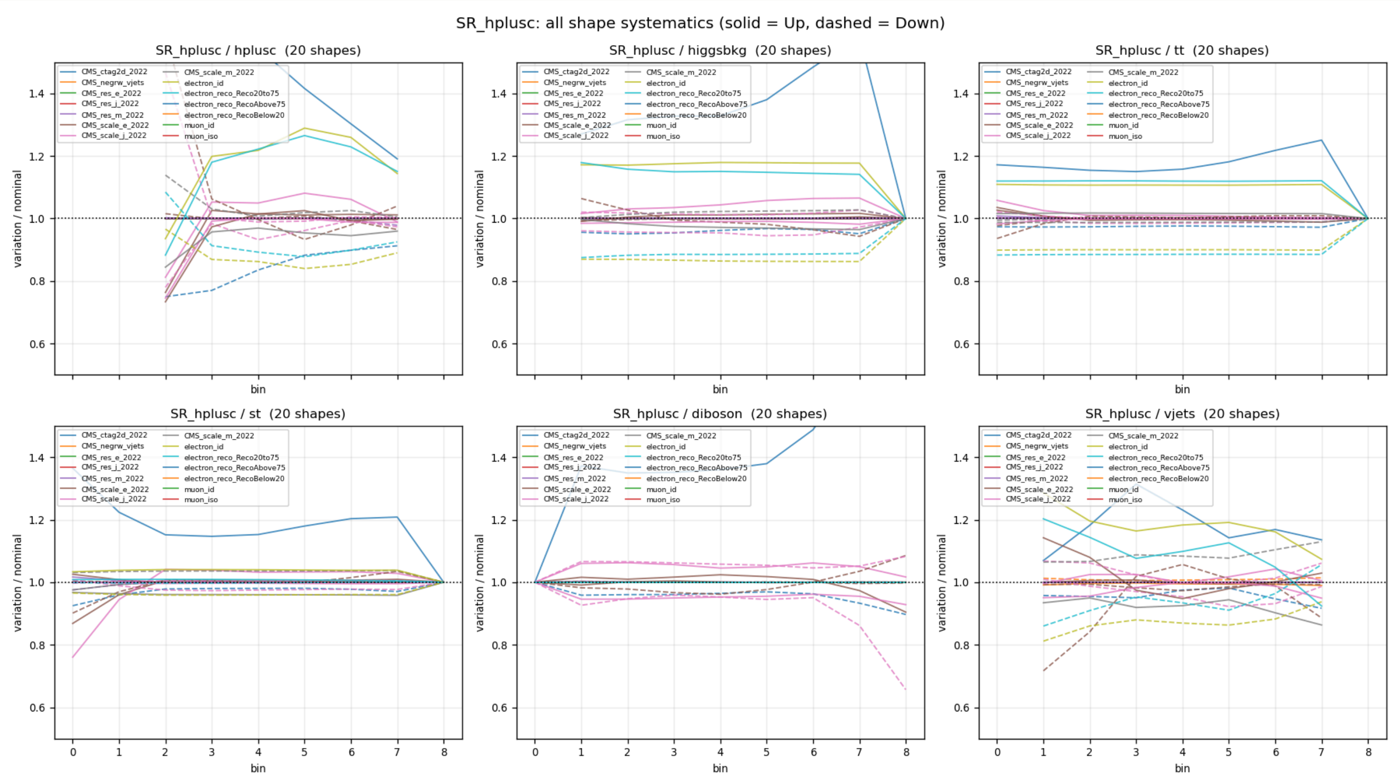
region	events	dominant
SR_hplusc	20,664	tt 82%
CR_tt	44,199	**tt 94%**
CR_vjets	9,214	**vjets 43%**
CR_higgsbkg / st / diboson	6.6k–20k	tt-rich

> CR_tt at **94% purity** pins the free-floating `rate_tt`, which covers 82% of the SR.

Templates in all six regions

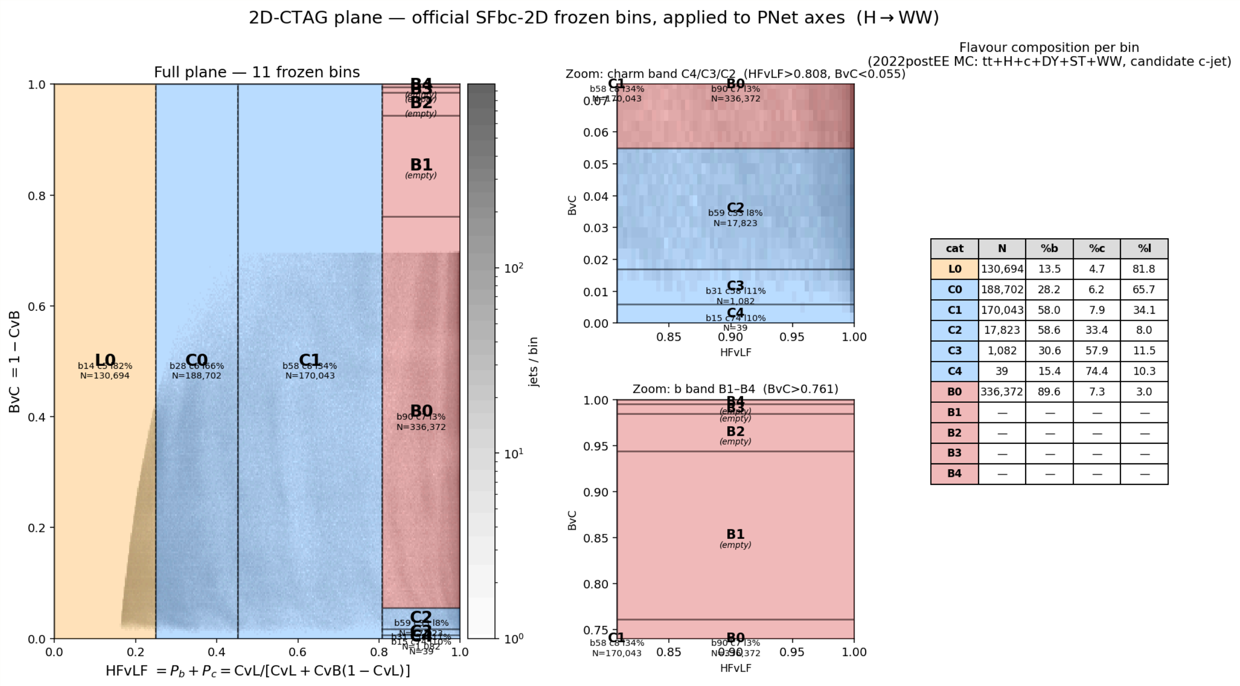


Signal region template



3 · c-tagging

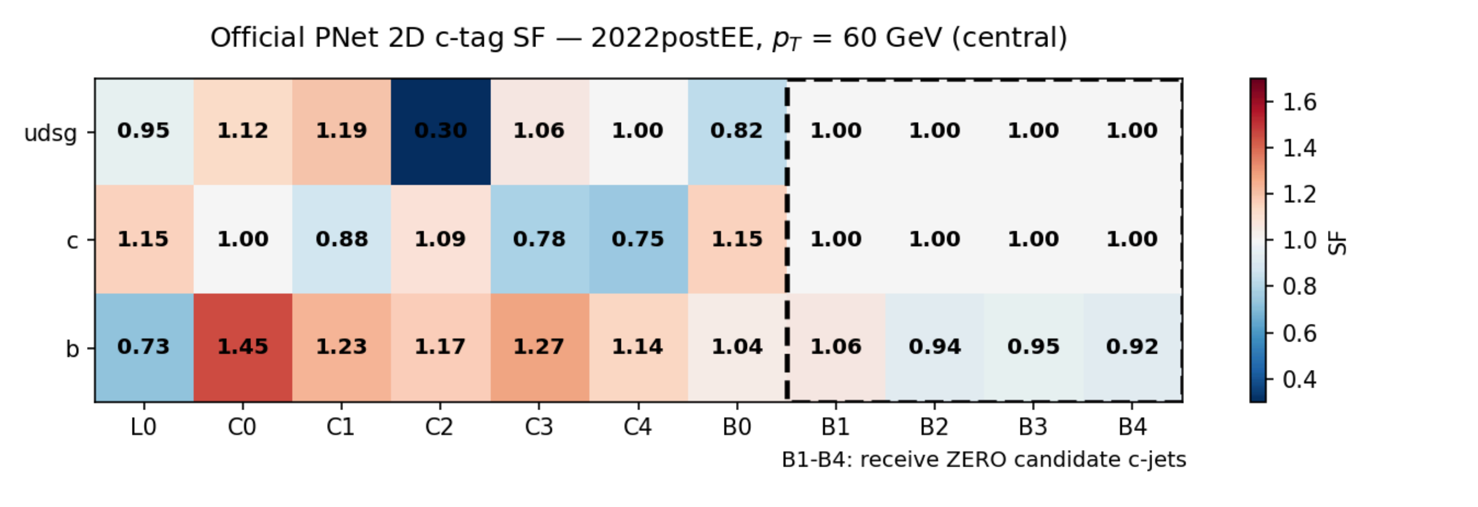
The 2D plane



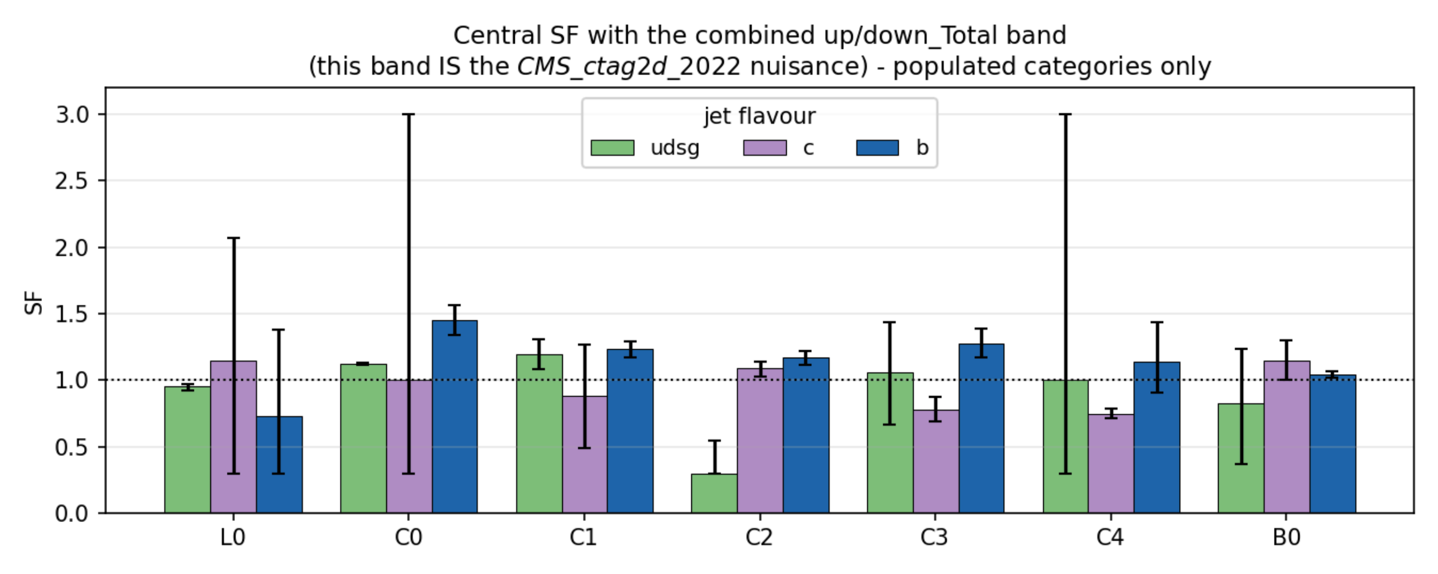
cat	N	%b	%c	%l
L0	130,694	13.5	4.7	81.8
C0	188,702	28.2	6.2	65.7
C1	170,043	58.0	7.9	34.1
C2	17,823	58.6	33.4	8.0
C3	1,082	30.6	57.9	11.5
C4	39	15.4	74.4	10.3
B0	336,372	89.6	7.3	3.0
B1	—	—	—	—
B2	—	—	—	—
B3	—	—	—	—
B4	—	—	—	—

11 categories `L0, C0–C4, B0–B4` spanning the whole CvL/CvB plane.

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ `CMS_ctag2d_2022`	**1164**

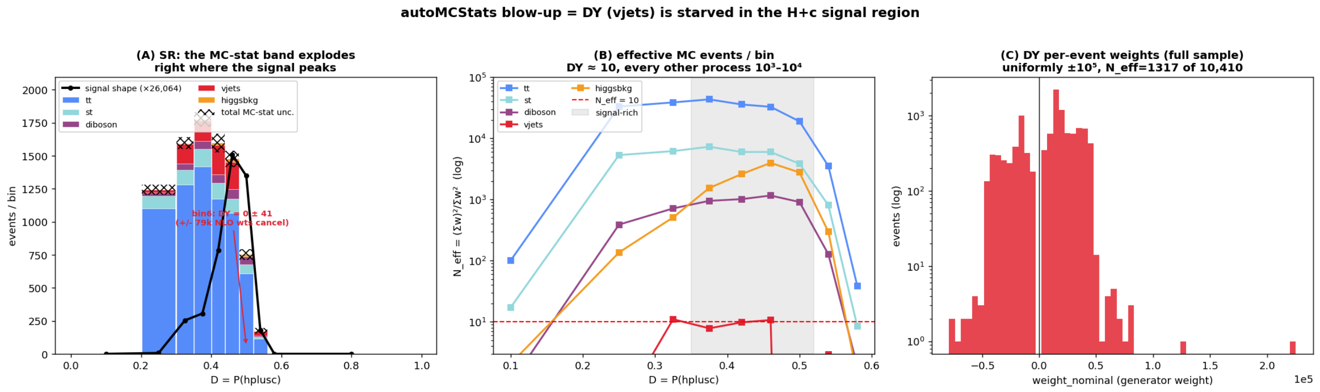
Applying the calibration **worsens** the limit by 14 units — as it must.

A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

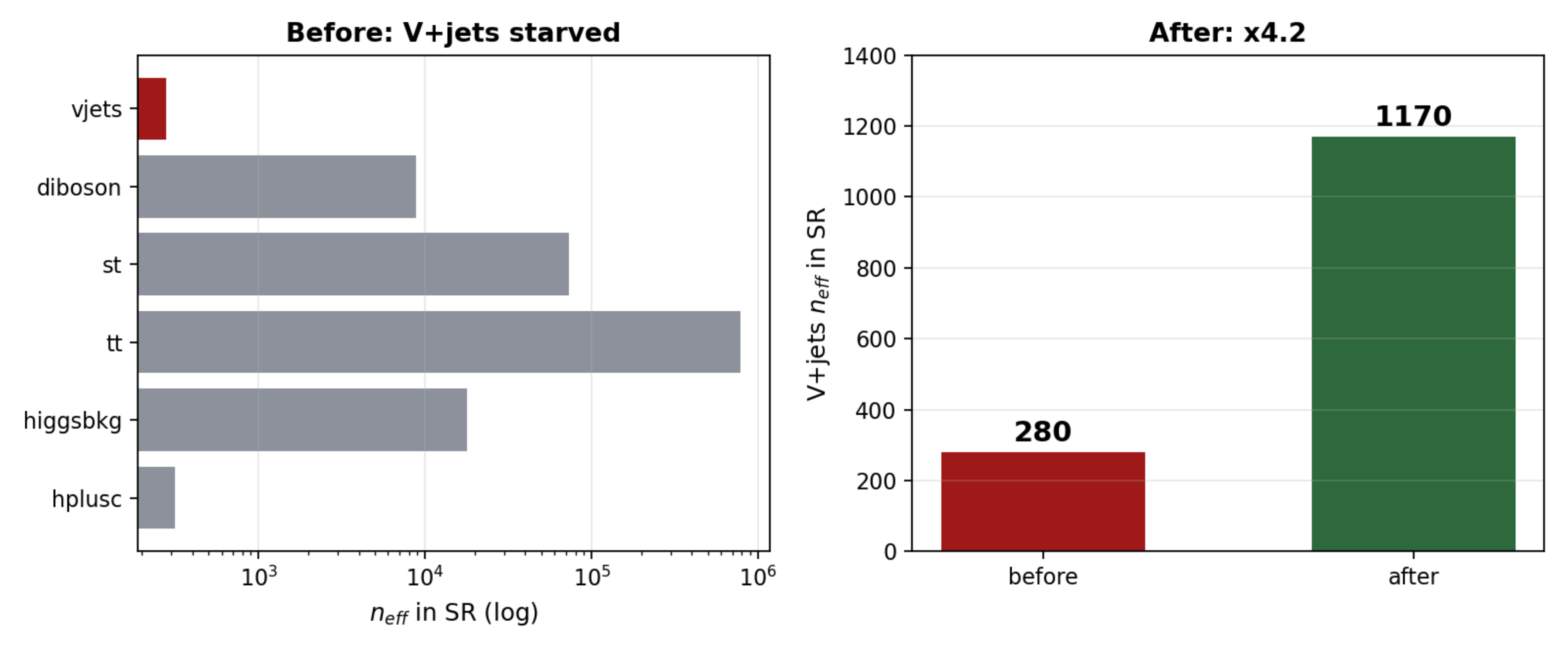
4 · Negative-weight reweighting

The problem



aMC@NLO weights are **signed** — the yield is a cancellation.

V+jets was starved exactly under the signal



$n_{eff} = (\sum w)^2 / \sum w^2$ — every other background sits at ≤1.1%.

The method

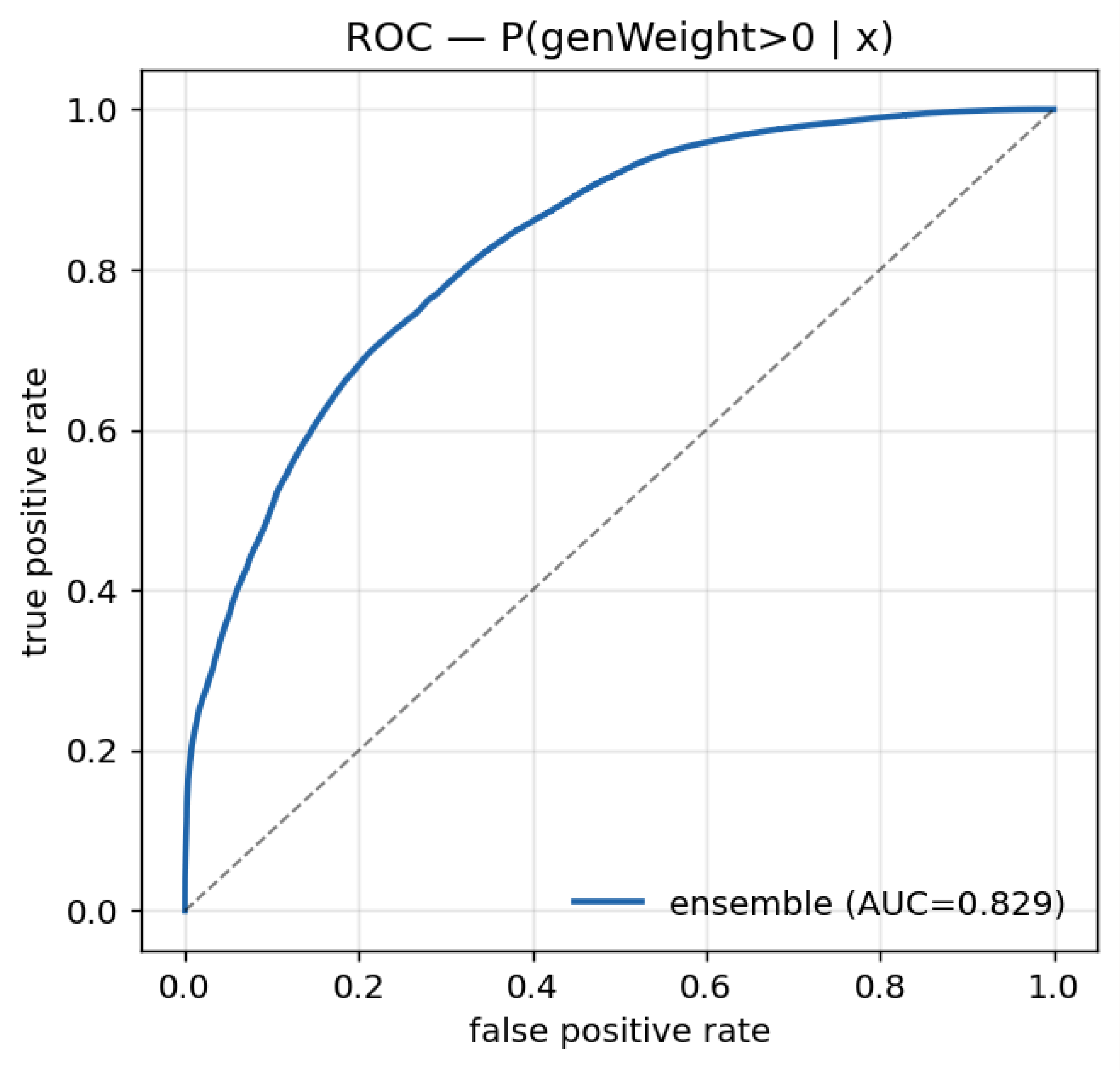
$$w \;\to\; |w| \cdot g(x) \cdot \mathrm{renorm}, \qquad g(x) = 2P_{+}(x) - 1$$

An algebraic identity, not an approximation — yield-preserving by construction.

- **generator-level features only** — P_{+} is a property of the generator
- **train loose, infer tight**, disjoint by construction from the eμ SR
- same phase space, disjoint events → interpolation, never extrapolation

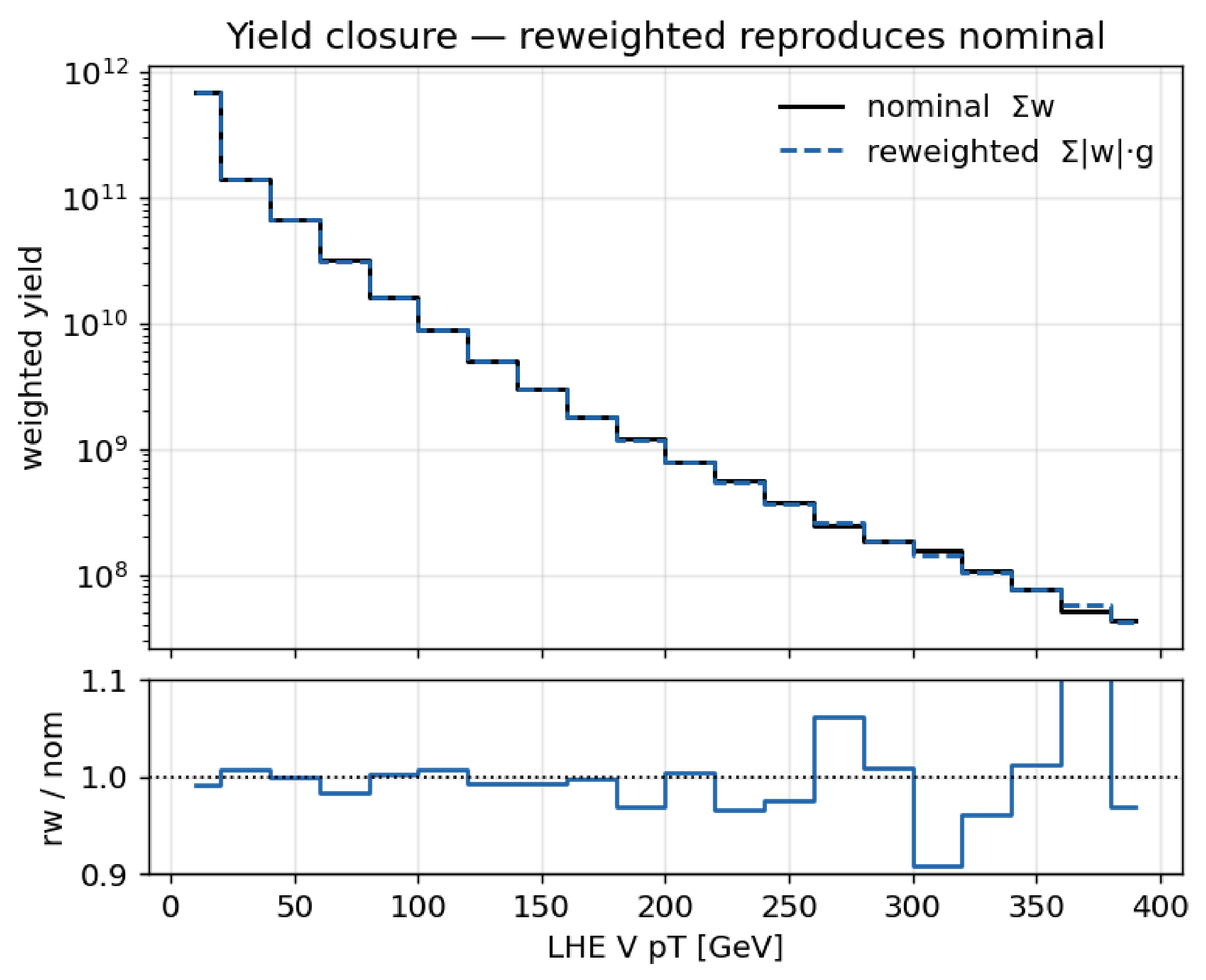
arXiv:2510.16217

Classifier performance



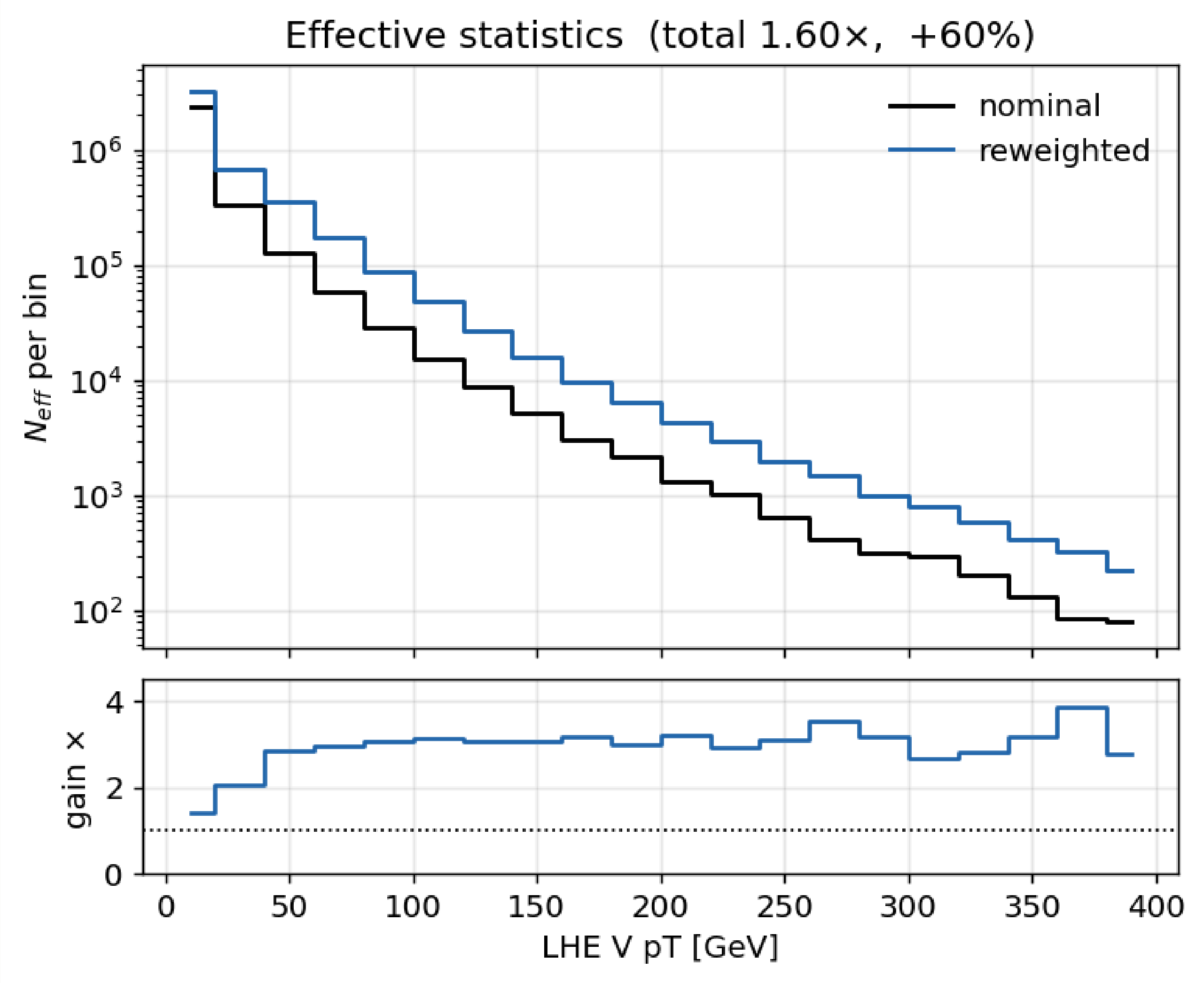
Ensemble **AUC 0.829** on an intrinsically stochastic target.

Closure



Training-region closure **0.994** on 9.8M events.

The gain



Method uncertainty `CMS_negrw_vjets` profiles to **0.0%** — negligible.

5 · W+jets jet-binned samples

Inclusive → jet-binned

AN-23-102 rejects the inclusive NLO sample: *large negative-weight fraction.*

sample	xsec (pb)	events	neg-w
0J	55,760	678M	10%
1J	9,529	523M	26%
2J	3,532	345M	35%
sum	**68,821**	**1,546M**	
inclusive	*67,710*	*282M*	*16%*

Cross sections from XSDB · sum within **+1.6%** of the inclusive.

Result

	before	after
expected UL	1160	**1034**
V+jets n_{eff} (SR)	280	**1170**
V+jets stat. error	5.98%	**2.92%**
V+jets rate	1508	1523

The rate barely moved while the statistical error halved — the signature of a statistics gain, not a normalisation change.

Was the gain really W+jets?

`tt` (+4.8%) and `st` (+3.8%) also moved between the two cards — both were re-merged from scratch. Checked per-process:

process	n_{eff} **ratio**
vjets (SR)	**4.18×**
vjets (CR_st)	**9.89×**
vjets (CR_tt)	**5.86×**
tt, st, diboson, higgsbkg	0.87 – 1.13×

> **The n_eff gain is confined to V+jets.** Other processes move by ≤13% in *both* directions — re-merge jitter, not a systematic shift. Both cards were also re-verified independently: **1160** and **1034**.

6 · Systematics and impacts

What is in the card

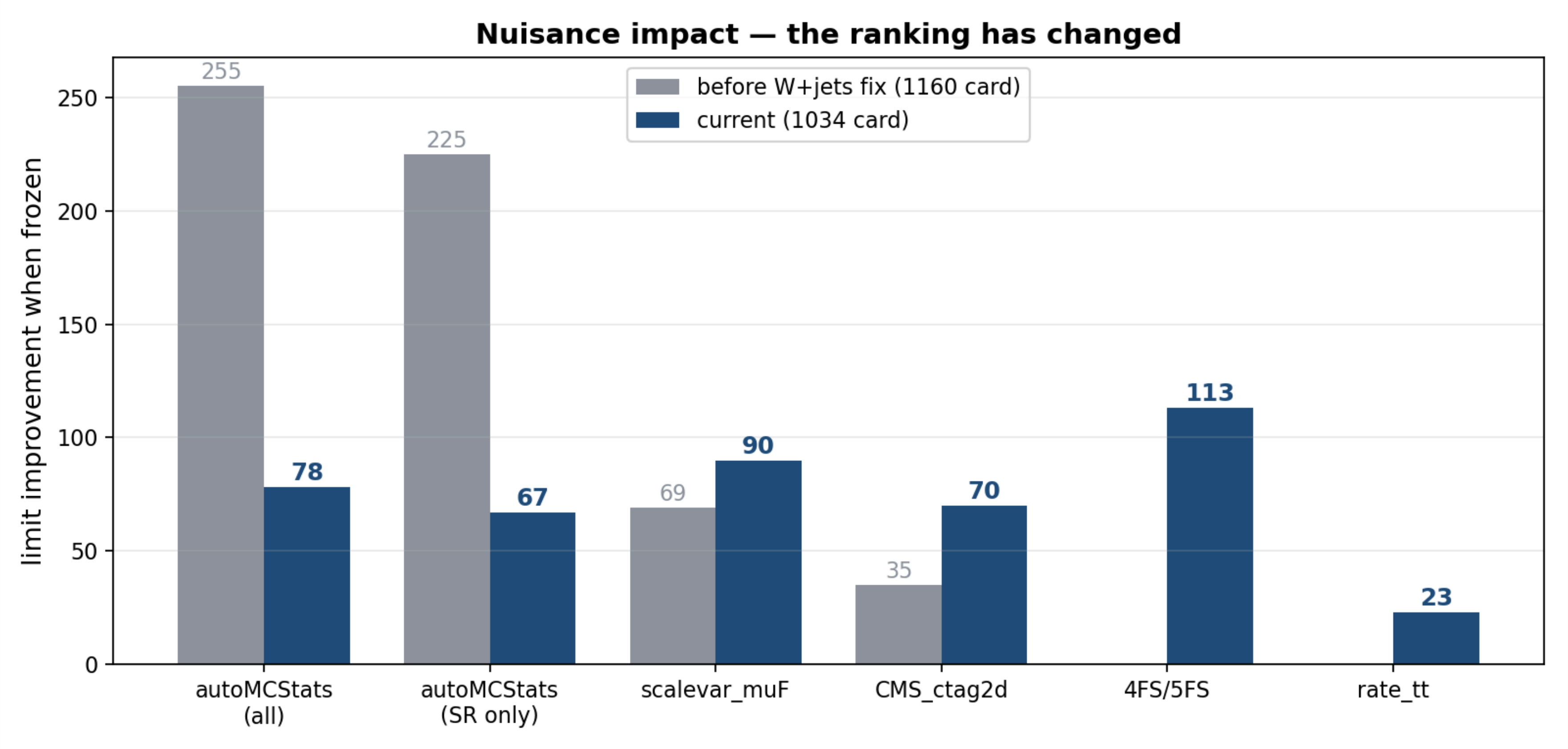
Shape (weight): pileup · `ps_isr/fsr` · `scalevar_muR/muF/muR_muF` · `muon_id/iso` · `electron_id` · `electron_reco` (3 p_T bins) · `CMS_ctag2d_2022` · `CMS_negrw_vjets`

Shape (object shift): JES · JER · electron scale/res · muon scale/res

Rate: lumi · `xsec_st/diboson/vjets/higgsbkg` · `BR_HtoWW` · `xsec_hplusc_PDF` · `xsec_hplusc_4FS_5FS` · `alphaS_PDF`

Plus: `rate_tt` free rateParam · autoMCStats (Barlow–Beeston)

Impacts — measured on the current card



The ranking has flipped

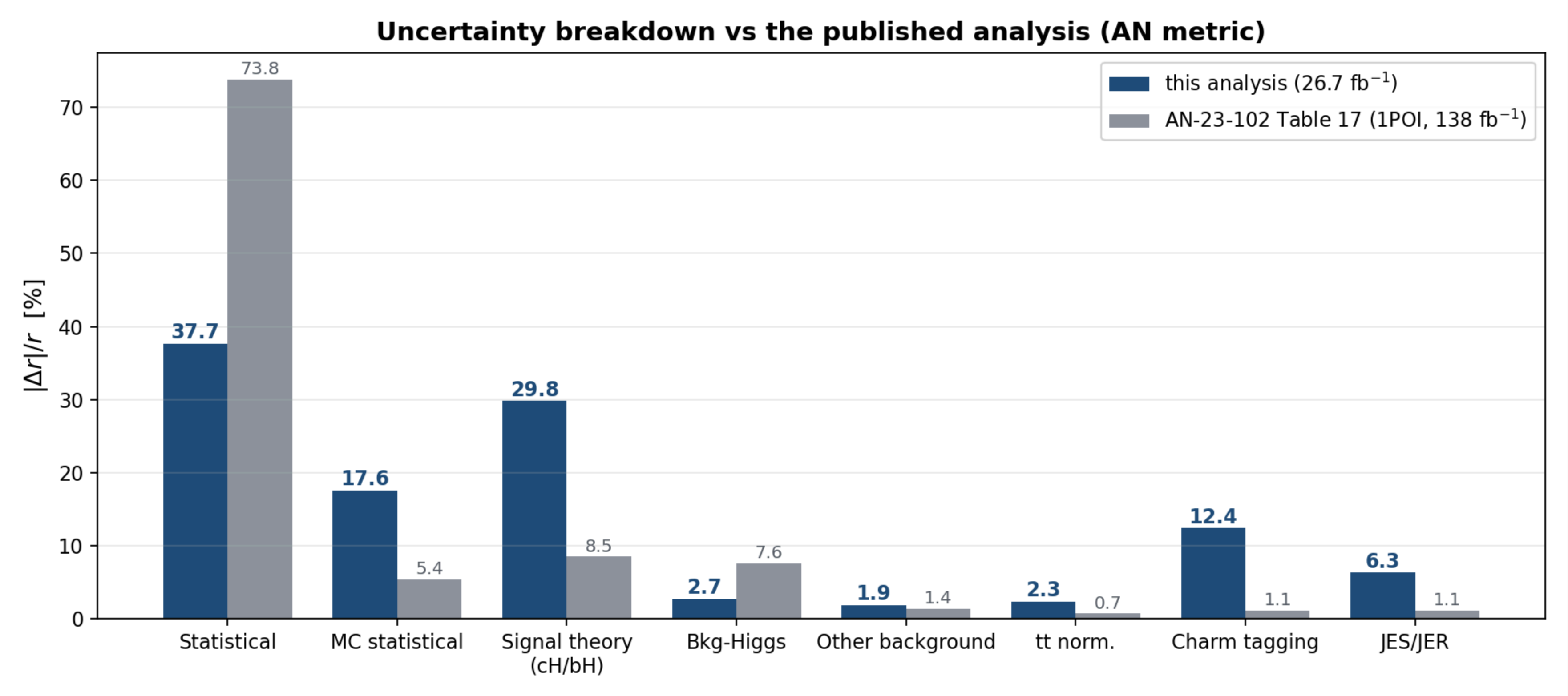
frozen	before	now
autoMCStats (all)	−255	**−78**
`xsec_hplusc_4FS_5FS`	—	**−113**
`scalevar_muF`	−69	**−90**
`CMS_ctag2d_2022`	−35	−70

> **MC statistics is no longer the leading systematic.** Signal theory is.

Ranked impacts (Asimov)

nuisance	impact on r
`scalevar_muF` / `scalevar_muR_muF`	153
`prop_bin` SR bins 8–9	122 / 98
`CMS_scale_j_2022` (JES)	94
`rate_tt`	87
`CMS_ctag2d_2022`	43

Breakdown vs the Run 2 analysis



Both on the AN's metric: $|\Delta r|/r = \sqrt{\sigma^2_{full} - \sigma^2_{frozen}}/r$

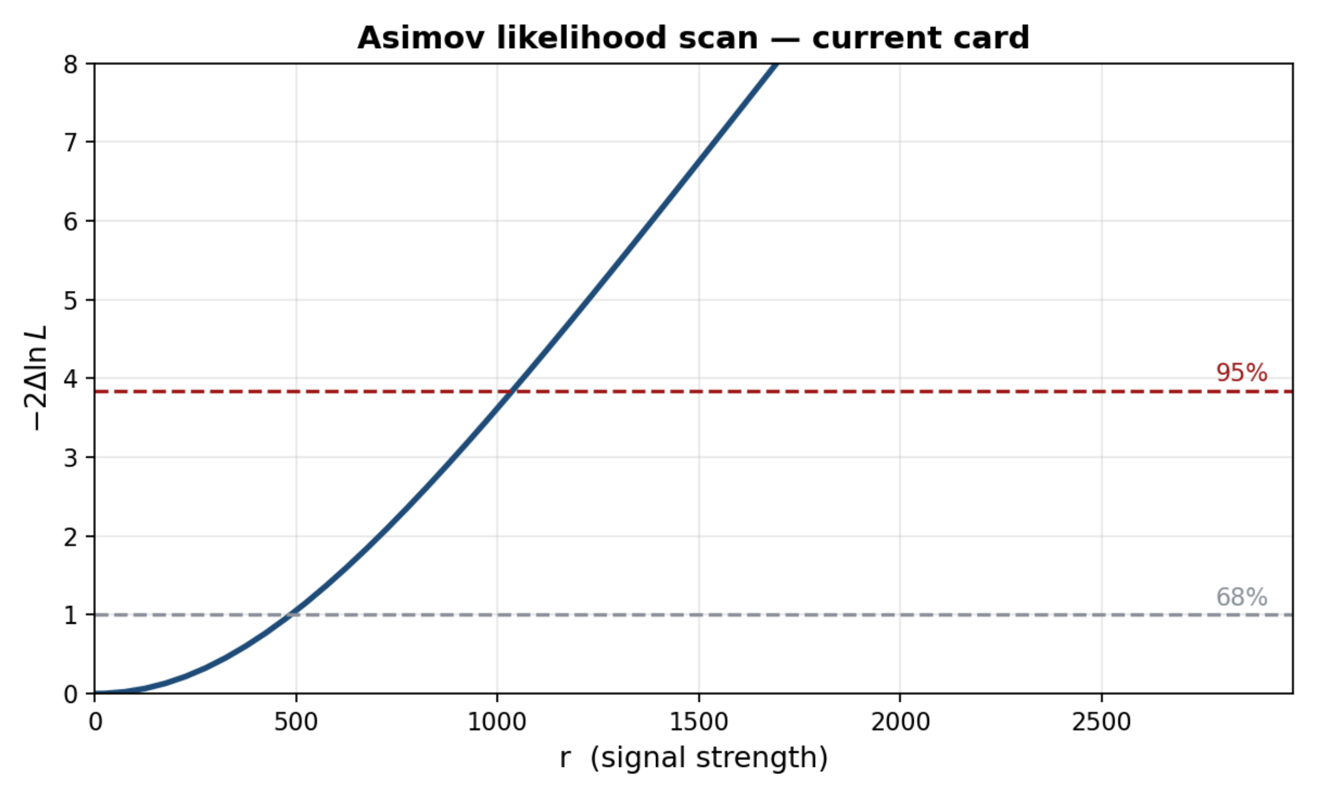
The comparison, term by term

group	ours	AN 1POI	ratio
Statistical	37.7%	73.8%	0.5×
Signal theory (cH/bH)	**29.8%**	8.5%	**3.5×**
MC statistical	**17.6%**	5.4%	**3.3×**
Charm tagging	12.4%	1.1%	11×
JES/JER	6.3%	1.1%	5.7×
tt normalization	2.3%	0.7%	3.3×
Bkg-Higgs	2.7%	7.6%	**0.4×**
Other background	1.9%	1.4%	1.4×

MC statistical was **28.7%** before the W+jets fix — now **17.6%**.

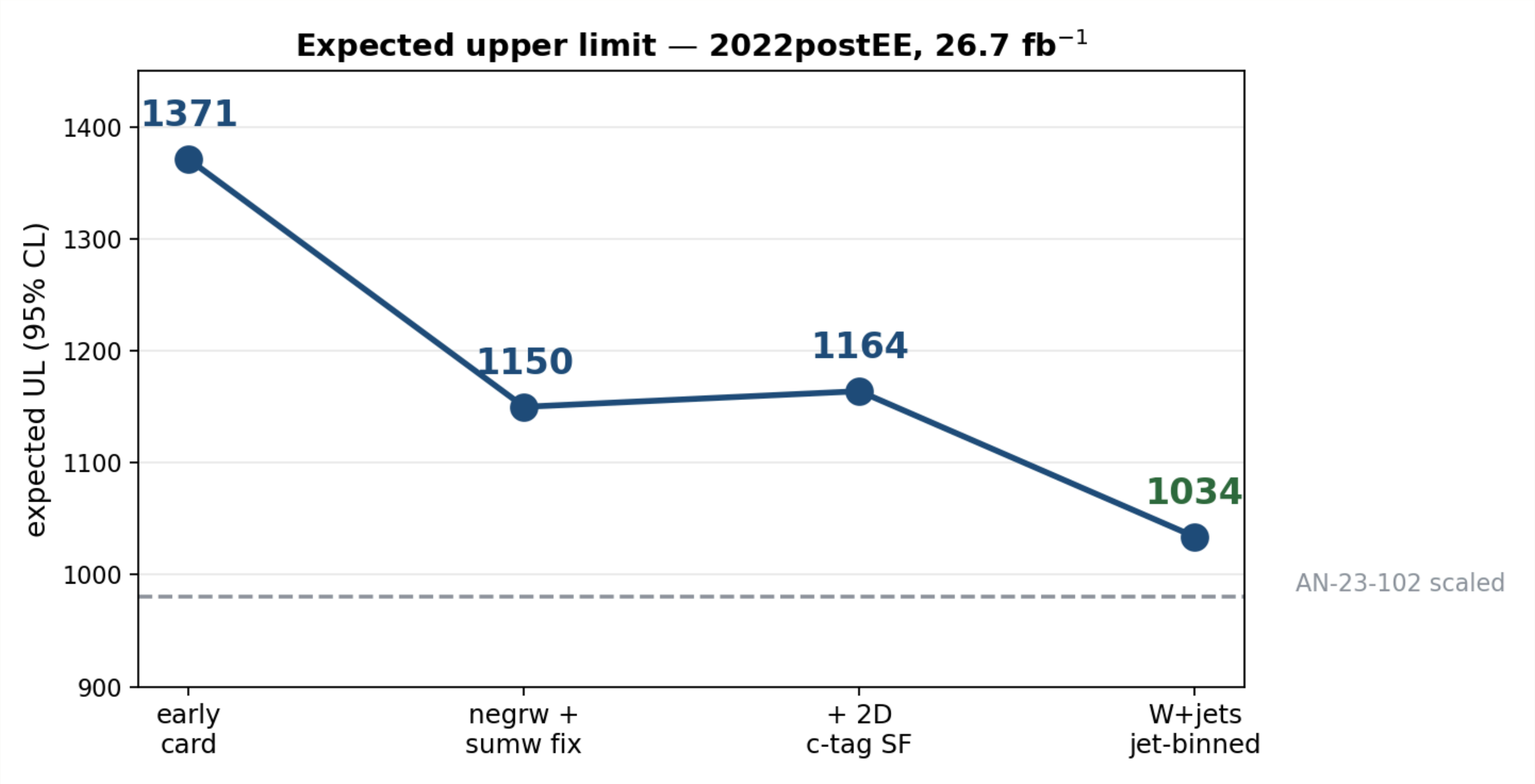
Components are **not orthogonal** — they do not sum to 100%. The AN's own column sums to 100.8% for the same reason.

Likelihood scan



7 · Status and outlook

The road so far



Where we stand

	value
AN-23-102 scaled to 26.7 fb^{-1}	980
this analysis	**1034**
our stat-only	676
AN stat-only scaled	~723

Our statistics-only limit already beats the AN's.

The remaining gap is entirely systematic.

Next steps

priority	item
1	Re-derive `xsec_hplusc_4FS_5FS` at 13.6 TeV — now the largest lever
2	Investigate `scalevar_muF`
3	Add missing `-ext` tt/st samples
4	DY $\rightarrow$ Z$\rightarrow\tau\tau$ filtered samples
5	Decorrelate `CMS_ctag2d_2022`
6	Enable trigger SFs

Missing systematics: **trigger SFs** (implemented, not enabled) and a **re-derived 4FS/5FS** term.

Summary

Expected UL 1371 → 1034

V+jets n_{eff} **280 → 1170**

The remaining gap is systematic, not statistical

Signal theory is now the leading term, not MC statistics